

The signed border mean

triple sum exact; pole not the carrier;
the bound sits at the circularity gate

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Abstract

r442 reduced the remaining RH path to a positive frequency of $q^\dagger < 1$ on dyadic ζ -anchors, with $q^\dagger = B_w^{-1} b^T (I - C)^{-1} b$. This note writes the block mean as an explicit triple sum over windows and signed border atoms, splits diagonal from off-diagonal and the almost-mode pole from the regular resolvent, and attempts a classical mean-value bound. The identities are theorems of finite linear algebra. The bound is *circular*: the load-bearing object is signed off-diagonal cancellation in the *regular* kernel, of strength matching a diagonal that already exceeds 40 at $k = 9$, against a window-dependent OP-resolvent kernel that is not a classical $k(\log n/n')$. That is the second precise circularity verdict of the path (after r399). No RH claim.

Status. Lemmas 1–2 are **SATZ**. Propositions 1–3 are machine-checked census or named refutations. The mean bound is *circular*, not open-untested: the missing signed strength is named in Section 5. Ausgang ZIRKULAER / TRIPLE_SUM_EXACT / POLE_NOT_CARRIER / DIAGONAL_OVERFLOW / OFFDIAG_LOADBEARING / DEAD_CHI_POLE_OVERSHOOT. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The triple sum

r420/r424 write b as the μ -OP coefficient of the signed border measure σ on atoms (t_a, σ_a) : $b = \sum_a \sigma_a \phi(t_a)$. Woodbury gives $q^\dagger = B_w^{-1} b^T (I - C)^{-1} b$ with $C = B^T B$. Expanding the quadratic form:

Lemma 1 (Triple sum). *On each window W_k*

$$q^\dagger(W_k) = \sum_{a,a'} \frac{\sigma_{k,a} \sigma_{k,a'}}{B_{w,k}} \langle \phi_k(t_a), (I - C_k)^{-1} \phi_k(t_{a'}) \rangle,$$

hence

$$\frac{1}{K} \sum_{k=1}^K q^\dagger(W_k) = \frac{1}{K} \sum_k \sum_{a,a'} \frac{\sigma_{k,a} \sigma_{k,a'}}{B_{w,k}} K_k(a, a'),$$

where $K_k(a, a') = \langle \phi_k(t_a), (I - C_k)^{-1} \phi_k(t_{a'}) \rangle$. Over $\mathbb{Q}$: two atoms $\phi = (1, 0)$ and $(1, 1)$ against $C = \text{diag}(1/2, 0)$ give $q = 9 = 5 + 4$. On w9 the reconstruction $b = \Phi \sigma$ holds at $1.4 \cdot 10^{-16}$ and matches $q^\dagger$ at 10^{-15} .

Selected anchors $a_k = 2^k$ are ζ -windows: the X -atoms include primes in dyadic ranges (von Mangoldt data), the Y -atoms are holes with opposite sign.

2 Diagonal versus off-diagonal

Write $q^\dagger = q_{\text{diag}} + q_{\text{off}}$ with $q_{\text{diag}} = B_w^{-1} \sum_a \sigma_a^2 K(a, a)$ the unsigned skeleton (signs drop).

Proposition 1 (Diagonal overflows). *Selected $k = 3, 4, 5, 6, 7, 9$:*

k	q_{diag}	q_{off}	$q^\dagger$	$q_{\text{diag}} < 1?$
3	0.515	+0.413	0.928	yes
4	0.482	+0.451	0.933	yes
5	0.892	-0.001	0.891	yes
6	0.835	+0.097	0.931	yes
7	2.272	-1.327	0.945	no
9	40.777	-39.815	0.962	no

Mean $q_{\text{diag}} = 7.629 > 1$. From $k = 7$ the signed off-diagonal is load-bearing: without it, $q^\dagger > 1$ already at $k = 7$.

A positive diagonal majorant therefore cannot prove $\frac{1}{K} \sum q^\dagger < 1$. This is the same family as the r442 unsigned envelopes ($|b| = 38.5$ on w9, $|\sigma| = 2.10$ at $k = 5$ and 87 at $k = 9$), now seen inside the signed expansion: even σ^2 on the kernel diagonal overflows. The arithmetic is the off-diagonal cancellation $q_{\text{off}} \approx -q_{\text{diag}} + O(1)$.

The pairwise phases $\log t_a - \log t_{a'}$ are the classical bilinear differences. On an 80-atom X -subsample of w9, $\text{corr}(K_{\text{off}}, \cos \Delta \log) = 0.026$: the kernel is *not* a cosine of the log-ratio.

3 Pole versus regular resolvent

Let $\lambda_{\text{max}}(C_k) < 1$ (true on every measured living and dead window: $n_{\geq 1} = 0$) with unit eigenvector ψ_{max} . As in r436,

$$(I - C)^{-1} = \frac{\psi_{\text{max}} \psi_{\text{max}}^T}{1 - \lambda_{\text{max}}} + R_{\text{reg}}.$$

Then $q^\dagger = q_{\text{pole}} + q_{\text{reg}}$ with $q_{\text{pole}} = B_w^{-1} |\langle b, \psi_{\text{max}} \rangle|^2 / (1 - \lambda_{\text{max}})$.

Lemma 2 (Spectral split). *Over $\mathbb{Q}$: $C = \text{diag}(1/2, 1/4)$, $b = (1, 1)$ give $q = 10/3 = 2 + 4/3$. On every selected window the split residual is $\leq 10^{-12}$.*

Proposition 2 (Pole is not the carrier). *Selected pole shares: 0.0003, 0.0007, 0.0031, 0.0009, 0.0005, 0.0001. Mean $q_{\text{pole}} = 8.6 \cdot 10^{-4}$. The regular part is $q^\dagger$ to 10^{-3} , and follows the same rise toward 1 (0.928 $\rightarrow$ 0.962). The C -gap $1 - \lambda_{\text{max}}$ shrinks as the r440 collar ($2.99 \cdot 10^{-3} \rightarrow 1.91 \cdot 10^{-8}$), but $\langle b, \psi_{\text{max}} \rangle$ shrinks faster: the almost-mode is nearly orthogonal to the border vector on living windows. A near-1 cluster grows (65 modes with gap $< 10^{-2}$ at $k = 9$) yet carries only 0.8 percent of $q^\dagger$. The living mean is a bulk regular quadratic form, not a pole-weighted prime sum.*

The hoped-for r436-style reduction (pole carries growth, remainder block-stable $\Rightarrow$ the mean is a pole-weighted Λ -sum) is therefore refuted as a strategy for the living mean. The remainder is not block-stable: it is the trending $q^\dagger$.

4 Dead χ : pole overshoot

Proposition 3 (Death certificate). *On all six dead χ windows, $q^\dagger > 1$, $q_{\text{reg}} < 1$, and $q_{\text{pole}} > 1 - q_{\text{reg}}$. CHI_3 -15: $q^\dagger = 1.023$, $q_{\text{pole}} = 0.038$, $q_{\text{reg}} = 0.985$. CHI_3 -39 has the largest pole (0.095). Living CHI_3 -9: $q_{\text{pole}} = 0.003$. Dropping the pole misclassifies every dead window as living.*

So the almost-mode *does* distinguish live from dead, but as a death spike, not as the carrier of the living mean. On MAIN (core-42) the risk is the opposite: q_{reg} itself sits at 0.9985 (kz16), with mean top-mode $4.5 \cdot 10^{-4}$. Two mechanisms, only one seen as death in the census.

5 Bound inventory and the gate

Each brick is typed. None closes.

- **Diagonal majorant** (unconditional). REFUTED: mean $q_{\text{diag}} = 7.63 > 1$, and $q_{\text{diag}} > 1$ pointwise from $k = 7$.
- **Pole-weighted prime sum** (would be PNT on the main term of $\langle \sigma, \psi_{\text{max}} \rangle$, RH-near on the oscillation). REFUTED as a reduction: pole share < 0.4 percent on Selected. PNT is idle.
- **Montgomery–Vaughan MVT** on $\sum_{a \neq a'} \sigma_a \sigma_{a'} K_k(a, a')$ (unconditional). The kernel K_k depends on the window through C_k and the OP of W_k . It is not a translation-invariant $k(\log n/n')$ (corr = 0.026 vs. cosine). MVT does not apply in the form that would bound a classical bilinear prime form. Even if forced, MVT gives a mean-square envelope of size $\asymp q_{\text{diag}}$, not cancellation to relative error $1/40$.
- **Cauchy–Schwarz** between block and atoms (unconditional). Loses the sign of q_{off} . At $k = 9$, $|q_{\text{off}}|/q_{\text{diag}} = 0.976$ (near-perfect cancellation). CS cannot see it.
- **Gallagher large sieve** (unconditional). Wants a Fourier multiplier in $t = \log$. The kernel is not one. Structurally inapplicable.
- **RH-quality bilinear estimate** on $\sum \Lambda(n)\Lambda(n') R_k^{\text{reg}}(n, n')$. This *would* give the missing cancellation. It is the circularity gate: an error term of that quality on signed von-Mangoldt correlations is RH-near (the zeros of ζ are those correlations). Forbidden as a mean-value carrier, as in r399.

The missing signed strength. A uniform bound

$$\left| \frac{1}{K} \sum_k \sum_{a \neq a'} \frac{\sigma_{k,a} \sigma_{k,a'}}{B_{w,k}} R_k^{\text{reg}}(a, a') \right|$$

of strength matching the diagonal growth (q_{diag} already 40 at $k = 9$), with $R_k^{\text{reg}} = (I - C_k)^{-1}$ minus the rank-one almost-mode, a window-dependent OP-resolvent kernel on dyadic ζ -anchors. This is the second precise circularity verdict of the path (r399: Weyl-energy decay at CD-transfer scale was RH-near *and* independently refuted by a growing census; here the census is living, the bound itself is circular).

r430 `exists_index_zero_of_block_mean_lt_one` is already proved and is not in question. The analytic input it needs is not supplied by any unconditional brick above.

6 Kills

- **Unsigned regression.** $|b|$ -envelope $w_9 = 38.49$; $|\sigma|$ at $k = 5, 9$ is 2.10 and 87.1 (r442 reproduced).
- **Drop-pole mutant.** $q_{\text{reg}} < 1$ on 6/6 dead χ : the mutant declares false life. On living windows drop-pole is invisible (pole share ~ 0).

- **Wrong block normalisation.** $q \cdot B_w = 7.82$ on w9, not a < 1 object.
- **Soft means (r440).** $\text{tr } G^\dagger$ is 50.9 living vs. 53.0 dead; $M_2 = 32.96$ vs. 39.01. Both $\gg$ threshold.
- **Scramble.** Unaugmented $\text{nneg}(R) = 21$.

Claim boundary

Nothing in this note is evidence for or against the Riemann hypothesis, in either direction. Lemmas 1–2 are finite linear algebra. The mean-value bound that would feed r430 sits at the circularity gate named in Section 5.

References

- [1] Round 399, `source_weyl_energy`: first circularity verdict (Weyl energy).
- [2] Round 436, `p2_determinant`: pole / regular template.
- [3] Round 440, `mean_tau_index`: collar gap; soft means blind.
- [4] Round 442, `block_mean`: $\kappa^\dagger = 1\{q^\dagger > 1\}$; unsigned majorant refuted.
- [5] `exists_index_zero_of_block_mean_lt_one`.